Florula Aramiensis

A Flora of the Arami Uplands

Volume I

Economic Trees

Edited by

Rotimi Alade

Arami Uplands Herbarium

2026

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Note on the data
The Arami Uplands are a fictional region. The species treated in this volume are real West African trees, and their descriptions follow the published literature cited in each treatment. All specimen records, collectors' numbers, localities and dates are synthetic and were constructed for demonstration. They must not be cited as records of real collections.

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Preface

The Arami Uplands have no flora of their own. Collectors who worked in the region over the past century sent their specimens to herbaria in three different countries, and the names under which those specimens were filed have since been revised many times. A forester, an extension officer or a student who wants to identify a tree in the Uplands today has to work from regional floras that treat the area as a footnote.

This volume begins to close that gap. It treats the economic trees of the Uplands: the timber species, the fruit and seed trees that feed households and markets, and the trees whose bark, leaves or gums are used in medicine and craft. These are the species that people in the region most often need to name correctly, because a wrong name on a seedling order, a timber permit or a conservation list has real consequences.

The treatments are arranged by family, eight in this volume. Each family opens with a short description and a key to its genera and species. Each species treatment gives the accepted name and its principal synonyms, local names, a description, notes on phenology and habitat, the uses recorded in the literature, and the specimens on which the treatment rests.

Later volumes will cover the shrubs, climbers and herbs of the Uplands. Corrections and additional records are welcome and should be sent to the Arami Uplands Herbarium.

R. A.
Arami Uplands Herbarium
September 2026

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How to use this Flora

Arrangement

Families are arranged alphabetically, and within each family the genera and species are also arranged alphabetically. The Key to Families (Chapter 3) leads to a family; the key at the start of each family chapter leads to a species.

Keys

The keys are dichotomous. Each couplet offers two contrasting leads, numbered **1a** and **1b**, **2a** and **2b**, and so on. Choose the lead that fits the specimen and follow the dotted line to its end, which gives either the number of the next couplet or a name. The short example below shows the layout only.

1a Leaves compound 2
1b Leaves simple ..... Key to Families
2a Leaves bipinnate; flowers in pendent red heads ..... *Parkia*
2b Leaves once pinnate; flowers in panicles ..... *Khaya*

Names

Scientific names are printed in italics and are followed by their author citation in roman type, for example *Milicia excelsa* (Welw.) C.C.Berg. Author abbreviations follow the standard forms listed in the Author Abbreviations appendix. Principal synonyms are given under each accepted name.

Specimen citations

Each specimen is cited by collector, collector's number and the herbarium in which it is kept, for example Alade 104 (AUH). Herbarium codes used in this volume are:

Code	Herbarium
AUH	Arami Uplands Herbarium
OKF	Okeri Forestry Research Institute Herbarium
SVN	Savanna Botanical Station Herbarium

Abbreviations

alt. altitude
c. about (*circa*)
fl. flowering
fr. fruiting
s.n. without number (*sine numero*)
st. sterile (neither flowers nor fruits)
syn. synonym

Chapter 1

Introduction

1.1 The Arami Uplands

The Arami Uplands occupy about 4,800 km^2 of dissected plateau between the Ireke River in the south and the Asiri River in the north. The land rises from about 350 m along the Ireke valley to 900 m on the Aramiwo Escarpment, the highest ground in the region. The main settlements are Okeri, the district headquarters and seat of the Okeri Forestry Research Institute, and Idofin, a market town on the northern plateau.

1.2 Climate

The Uplands have two wet seasons, from April to July and from September to October, separated by a short dry spell in August. The long dry season runs from November to March and is broken by the dry, dusty harmattan wind from the north. Mean annual rainfall falls from about 1,600 mm in the Ireke valley to about 1,200 mm on the northern plateau.

1.3 Vegetation

The region lies in the transition between the Guineo-Congolian rain forest and the Sudanian savanna woodlands described by White (1983). Three broad vegetation types are recognised in this Flora.

Lowland semi-deciduous forest covers the southern slopes below about 500 m. Its canopy trees include *Khaya grandifoliola*, *Entandrophragma cylindricum*, *Milicia excelsa* and *Terminalia superba*.

Forest–savanna mosaic occupies the middle slopes between about 500 and 700 m, where gallery forest along streams alternates with grassland that burns most years.

Guinea savanna woodland covers the northern plateau above about 700 m. Here the fire-tolerant *Vitellaria paradoxa*, *Parkia biglobosa* and *Pterocarpus erinaceus* are characteristic, and many of them are kept in farmland because of the value of their fruits and seeds.

1.4 History of botanical collecting

The first substantial collections from the Uplands were made by E. M. Harding between 1934 and 1938, during forest reserve surveys around Okeri. J. A. Olaniyan collected widely on the northern plateau between 1961 and 1967 for the Savanna Botanical Station. The most recent collections were made by the editor between 2019 and 2025 for this Flora. Specimens from all three periods are cited in the species treatments.

1.5 Scope and sources

This volume treats the economic trees of the Uplands. Descriptions draw on the regional floras of Hutchinson and Dalziel (1954), Aubréville (1959), Keay (1989) and Aké Assi (2001), and on the field guides of Arbonnier (2004) and Hawthorne and Jongkind (2006). Information on uses follows Burkill (1985), Louppe et al. (2008) and Orwa et al. (2009). Nomenclature follows the *Shenzhen Code* (Turland et al., 2018), and author abbreviations follow Brummitt and Powell (1992).

Chapter 2

Materials and Methods

2.1 Data files

The Flora rests on four tab-separated data files in the `data` folder of this project. They are the single source for every count, date and locality printed in this volume.

`data/species.tsv`
: The checklist: one row for each of the 18 species, with its family, author citation, status (native or cultivated), the vegetation types in which it occurs, and its local and trade names.

`data/localities.tsv`
: The 12 collecting localities, with grid square, altitude and vegetation type. It matches the Gazetteer in Appendix B row for row.

`data/specimens.tsv`
: The 126 specimens on which the treatments rest, with collector, collector's number, date, locality, altitude, herbarium and phenological state.

`data/phenology.tsv`
: For each species and month, whether the species is recorded in flower (fl), in fruit (fr), both, or neither.

2.2 Validation of the specimen records

Every specimen record was checked against six rules before it was accepted.

1. The altitude equals the altitude of the locality in `localities.tsv`.

2. According to `species.tsv`, the species occurs in the vegetation type of the locality.

3. The date falls within the collector's working period: Harding 1934–1938, Olaniyan 1961–1967, Alade 2019–2025.

4. The herbarium is the one that holds that collector's material: Harding in OKF, Olaniyan in SVN, Alade in AUH.

5. A specimen in flower or fruit was collected in a month in which `phenology.tsv` records that state; a sterile specimen was collected in a month with neither.

6. Each collector's numbers increase with date and are never repeated.

2.3 Summary of the specimen data

Table 2.1 gives the number of species and specimens in each family, and Table 2.2 gives the number of specimens from each collector.

Table 2.1: Species and specimens by family.

Family	Species	Specimens
Anacardiaceae	2	14
Combretaceae	2	14
Fabaceae	3	21
Irvingiaceae	1	7
Malvaceae	3	21
Meliaceae	3	21
Moraceae	2	14
Sapotaceae	2	14
Total	18	126

Table 2.2: Specimens by collector and phenological state.

Collector	Herbarium	Flowering	Fruiting	Sterile	Total
Harding	OKF	7	30	9	46
Olaniyan	SVN	6	24	6	36
Alade	AUH	10	22	12	44
Total		23	76	27	126

Chapter 3

Key to Families

This key covers only the eight families treated in this volume. Where a family keys out twice, the genus reached at that lead is given in brackets.

1a Leaves compound 2

1b Leaves simple 5

2a Leaves digitately compound, the leaflets all arising from the tip of the petiole ..... Malvaceae (*Adansonia*, *Ceiba*)

2b Leaves pinnate or bipinnate 3

3a Fruit a woody capsule splitting into 4 or 5 valves; seeds winged ..... Meliaceae

3b Fruit not a valved capsule 4

4a Leaves once pinnate with a terminal leaflet; crushed leaves with a resinous smell; fruit a fleshy drupe ..... Anacardiaceae (*Spondias*)

4b Leaves bipinnate, or pinnate with a swollen pulvinus at the base of the petiole; fruit a pod or a flat winged disc ..... Fabaceae

5a Slash exuding white latex 6

5b Slash without white latex 7

6a Stipules present, leaving a scar round the twig; flowers small, crowded in spikes or heads Moraceae

6b Stipules absent; leaves crowded at the ends of the twigs; fruit a berry with large glossy seeds Sapotaceae

7a Terminal bud enclosed in a long conical stipule that leaves a ring scar; fruit a large drupe Irvingiaceae

7b Terminal bud not enclosed in a conical stipule 8

8a Branches in tiers, with the leaves crowded at their ends; fruit dry and winged ..... Combretaceae

8b Branches not in tiers; fruit not winged 9

9a Fruit a kidney-shaped nut borne on a swollen, fleshy stalk ..... Anacardiaceae (*Anacardium*)

9b Fruit of 4 or 5 spreading follicles with large seeds; petiole swollen at both ends Malvaceae (*Cola*)

Chapter 4

Anacardiaceae

Cashew family

Trees or shrubs with resin canals in the bark, so that the slash often smells of turpentine or mango. Leaves alternate, simple or pinnate, without stipules. Flowers small, usually 5-merous, in terminal or axillary panicles. Fruit usually a drupe. Two species in the Arami Uplands, one of them cultivated.

Key to the species

1a Leaves simple, obovate, leathery; fruit a kidney-shaped nut on a swollen fleshy stalk *Anacardium occidentale*

1b Leaves pinnate with a terminal leaflet; fruit an ovoid yellow drupe ..... *Spondias mombin*

4.1 *Anacardium occidentale* L.

Local and trade names. Yoruba: Kaju. Trade: Cashew.

Status. Cultivated. Introduced from tropical America and now widely planted in farms and plantations.

Description. Evergreen tree to 12 m with a low, spreading crown. Leaves simple, alternate, obovate, leathery, 8–20 × 6–12 cm, rounded or notched at the apex. Flowers small, fragrant, pale with pink or red streaks, in terminal panicles. Fruit a kidney-shaped nut 2–3 cm long, borne on a swollen, pear-shaped, red or yellow fleshy stalk (the cashew apple).

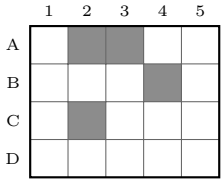
Phenology. Flowering December to March; fruiting February to May.

Habitat. Farms and plantations in the forest–savanna mosaic and Guinea savanna woodland.

Uses. The roasted kernel is eaten and sold, and the cashew apple is eaten fresh or pressed for juice. The oil in the shell is caustic and is used in industry (Burkill, 1985; Orwa et al., 2009).

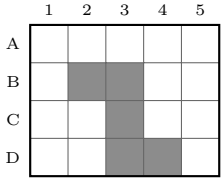
Specimens examined. Agbede (C2), 540 m, st., 7 Jul 1936, Harding 1534 (OKF); Agbede (C2), 540 m, fr., 6 May 1937, Harding 1659 (OKF); Agbede (C2), 540 m, fr., 19 Apr 1938, Harding 1960 (OKF); Oke-Asa farmland (B4), 690 m, st., 5 Jun 1965, Olaniyan 745 (SVN); Idofin (A3), 760 m, fr., 13 Apr 1967, Olaniyan 938 (SVN); Asiri Crossing (A2), 740 m, fl., 6 Jan 2019, Alade 133 (AUH); Asiri Crossing (A2), 740 m, fr., 8 Feb 2025, Alade 978 (AUH).

Distribution. Grid squares A2, A3, B4, C2.



4.2 *Spondias mombin* L.

Synonyms. *Spondias lutea* L.
Local and trade names. Yoruba: Iyeye. Trade: Hog plum.
Status. Native.
Description. Deciduous tree to 25 m with thick, corky, deeply fissured bark. Leaves imparipinnate with 5–10 pairs of leaflets; leaflets with an unequal base and a long-pointed apex. Flowers small, white or cream, in large terminal panicles. Fruit an ovoid drupe 3–4 cm long, yellow when ripe, with an acid, aromatic pulp.
Phenology. Flowering February to April; fruiting June to September.
Habitat. Lowland semi-deciduous forest and forest–savanna mosaic; common around villages, where it is planted from cuttings.
Uses. The fruit is eaten fresh. Large cuttings root readily and are used as living fence posts (Keay, 1989; Burkill, 1985).
Specimens examined. Olowo Stream (D3), 440 m, fr., 7 Jul 1934, Harding 1173 (OKF); Ireke Ferry (D4), 360 m, st., 15 Dec 1934, Harding 1260 (OKF); Igbo-Oba Forest Reserve (C3), 470 m, fr., 15 Jun 1937, Harding 1710 (OKF); Ewe Stream gallery forest (B3), 580 m, st., 27 Jan 1961, Olaniyan 262 (SVN); Aramiwo Foot (B2), 640 m, fr., 19 Sep 1966, Olaniyan 853 (SVN); Olowo Stream (D3), 440 m, st., 23 May 2022, Alade 611 (AUH); Olowo Stream (D3), 440 m, fr., 9 Sep 2024, Alade 862 (AUH).
Distribution. Grid squares B2, B3, C3, D3, D4.



Chapter 5

Combretaceae

Terminalia family

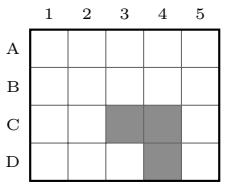
Trees, shrubs or climbers. Leaves simple and entire, alternate or opposite, in *Terminalia* crowded at the ends of short shoots on tiered branches. Flowers small, in spikes or heads. Fruit dry and indehiscent, usually with two to five wings. Two species of *Terminalia* in the Arami Uplands, both important timber trees.

Key to the species

1a Bark dark and deeply fissured; buttresses small; fruit longer than broad .... *Terminalia ivorensis*
1b Bark pale grey and smooth; tall plank buttresses; fruit much broader than long *Terminalia superba*

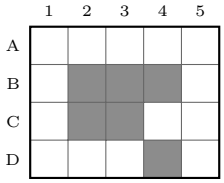
5.1 *Terminalia ivorensis* A.Chev.

Local and trade names. Yoruba: Idigbo. Trade: Idigbo.
Status. Native.
Description. Deciduous tree to 45 m with a straight, cylindrical bole and small buttresses. Bark dark grey to black, deeply fissured. Branches in tiers. Leaves crowded at the ends of the shoots, elliptic to obovate, 5–10 cm long. Flowers small, cream, in spikes. Fruit with two wings, oblong, longer than broad.
Phenology. Flowering February to April; fruiting April to June.
Habitat. Lowland semi-deciduous forest.
Uses. A light, easily worked timber used for joinery, window frames and furniture; also planted in forest plantations (Louppe et al., 2008).
Specimens examined. Igbo-Oba Forest Reserve (C3), 470 m, st., 3 Sep 1935, Harding 1378 (OKF); Ireke Ferry (D4), 360 m, fr., 15 Jun 1936, Harding 1483 (OKF); Igbo-Oba Forest Reserve (C3), 470 m, fr., 16 May 1937, Harding 1696 (OKF); Ireke Ferry (D4), 360 m, fr., 11 Apr 1938, Harding 1892 (OKF); Igbo-Oba Forest Reserve (C3), 470 m, st., 8 Dec 2020, Alade 399 (AUH); Ireke Ferry (D4), 360 m, st., 11 Jan 2021, Alade 435 (AUH); Okeri (C4), 420 m, st., 22 Aug 2023, Alade 806 (AUH).
Distribution. Grid squares C3, C4, D4.



5.2 *Terminalia superba* Engl. & Diels

Local and trade names. Yoruba: Afara. Trade: Afara.
Status. Native.
Description. Deciduous tree to 50 m with tall plank buttresses. Bark smooth, pale grey. Branches in tiers. Leaves crowded at the ends of the shoots, obovate, 6–12 cm long. Flowers small, greenish-white, in spikes. Fruit with two wings, much broader than long.
Phenology. Flowering February to April; fruiting May to August.
Habitat. Lowland semi-deciduous forest and forest–savanna mosaic, often in secondary forest.
Uses. A pale timber used for plywood, furniture and interior work, and widely planted in forest plantations (Keay, 1989; Louppe et al., 2008).
Specimens examined. Agbede (C2), 540 m, fr., 20 Jun 1934, Harding 1139 (OKF); Ireke Ferry (D4), 360 m, fr., 16 May 1937, Harding 1703 (OKF); Igbo-Oba Forest Reserve (C3), 470 m, fr., 23 Aug 1938, Harding 1992 (OKF); Ewe Stream gallery forest (B3), 580 m, fr., 17 Jun 1962, Olaniyan 413 (SVN); Aramiwo Foot (B2), 640 m, fr., 22 Jun 1966, Olaniyan 805 (SVN); Ewe Stream gallery forest (B3), 580 m, fr., 26 Aug 2021, Alade 544 (AUH); Oke-Asa farmland (B4), 690 m, fr., 5 Jun 2023, Alade 777 (AUH).
Distribution. Grid squares B2, B3, B4, C2, C3, D4.



Chapter 6

Fabaceae

Pea family

Trees, shrubs, climbers or herbs. Leaves alternate, usually pinnate or bipinnate, with stipules and with a swollen pulvinus at the base of the petiole. Flowers regular or irregular, often showy. Fruit a pod (legume), sometimes indehiscent and winged. Three species in the Arami Uplands, all trees of the forest–savanna mosaic and savanna.

Key to the species

1a Leaves bipinnate; flowers in pendent red heads on long stalks ..... *Parkia biglobosa*

1b Leaves once pinnate 2

2a Leaflets in 3–5 opposite pairs, without a terminal leaflet; fruit a thick woody pod; seeds black with an orange aril ..... *Afzelia africana*

2b Leaflets alternate, with a terminal leaflet; fruit a flat, round, winged disc, bristly over the seed *Pterocarpus erinaceus*

6.1 *Afzelia africana* Sm. ex Pers.

Local and trade names. Yoruba: Apa; Hausa: Kawo. Trade: Doussie.

Status. Native.

Description. Deciduous tree to 30 m, smaller in savanna, with a spreading crown. Bark grey, flaking in patches. Leaves paripinnate with 3–5 pairs of opposite, ovate-elliptic leaflets 5–10 cm long. Flowers with a single large petal, white streaked with red. Pod woody, flattened, 12–17 cm long, black when ripe. Seeds black, each with a bright orange aril.

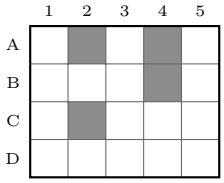
Phenology. Flowering March to May; fruiting November to February.

Habitat. Forest–savanna mosaic and Guinea savanna woodland.

Uses. A heavy, durable timber used for flooring, joinery and boat-building. The leaves are lopped for livestock fodder (Arbonnier, 2004; Louppe et al., 2008).

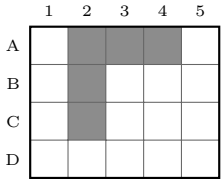
Specimens examined. Agbede (C2), 540 m, fl., 23 May 1935, Harding 1375 (OKF); Agbede (C2), 540 m, st., 18 Aug 1936, Harding 1558 (OKF); Agbede (C2), 540 m, st., 19 Jun 1937, Harding 1743 (OKF); Okuta Hill (A4), 820 m, fr., 27 Feb 1962, Olaniyan 385 (SVN); Oke-Asa farmland (B4), 690 m, st., 21 Jul 1963, Olaniyan 646 (SVN); Okuta Hill (A4), 820 m, fl., 11 May 2023, Alade 748 (AUH); Asiri Crossing (A2), 740 m, fr., 11 Nov 2024, Alade 901 (AUH).

Distribution. Grid squares A2, A4, B4, C2.



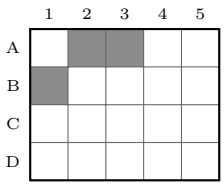
6.2 *Parkia biglobosa* (Jacq.) R.Br. ex G.Don

Local and trade names. Yoruba: Igba; Hausa: Dorowa. Trade: African locust bean.
Status. Native.
Description. Deciduous tree to 20 m with a wide, spreading crown. Bark thick, dark grey-brown, fissured. Leaves bipinnate with many pairs of pinnae, each bearing numerous small leaflets. Flowers in dense, pendent, red spherical heads on long stalks. Pods in clusters, 20–30 cm long, brown, the seeds embedded in a yellow, mealy pulp.
Phenology. Flowering December to February; fruiting March to May.
Habitat. Forest–savanna mosaic and Guinea savanna woodland; usually kept when land is cleared for farming.
Uses. The fermented seeds are made into a widely used condiment (Yoruba *iru*, Hausa *daddawa*). The sweet yellow pulp is eaten (Arbonnier, 2004; Orwa et al., 2009).
Specimens examined. Agbede (C2), 540 m, fr., 17 Apr 1934, Harding 1072 (OKF); Agbede (C2), 540 m, fl., 12 Feb 1935, Harding 1319 (OKF); Agbede (C2), 540 m, fr., 12 May 1936, Harding 1431 (OKF); Asiri Crossing (A2), 740 m, st., 3 Jul 1962, Olaniyan 429 (SVN); Idofin (A3), 760 m, fr., 9 Mar 1963, Olaniyan 556 (SVN); Okuta Hill (A4), 820 m, fl., 6 Feb 2020, Alade 256 (AUH); Aramiwo Foot (B2), 640 m, fl., 20 Jan 2023, Alade 647 (AUH).
Distribution. Grid squares A2, A3, A4, B2, C2.



6.3 *Pterocarpus erinaceus* Poir.

Local and trade names. Hausa: Madobiya. Trade: African rosewood.
Status. Native.
Description. Deciduous tree to 15 m. Bark dark grey and scaly, exuding a red gum when cut. Leaves imparipinnate with alternate leaflets. Flowers yellow, in panicles, appearing while the tree is leafless. Fruit a flat, round, winged disc 4–7 cm across, bristly over the seed.
Phenology. Flowering December to February; fruiting February to April.
Habitat. Guinea savanna woodland.
Uses. The red-brown timber is used for furniture and carving, and the leaves are lopped for fodder. Heavy logging for export has made it scarce in much of its range (Arbonnier, 2004; Orwa et al., 2009).
Specimens examined. Asiri Crossing (A2), 740 m, fl., 4 Feb 1962, Olaniyan 341 (SVN); Asiri Crossing (A2), 740 m, fr., 27 Feb 1963, Olaniyan 529 (SVN); Asiri Crossing (A2), 740 m, fr., 17 Apr 1965, Olaniyan 710 (SVN); Idofin (A3), 760 m, fr., 25 Apr 1967, Olaniyan 958 (SVN); Idofin (A3), 760 m, st., 15 Jun 2019, Alade 212 (AUH); Aramiwo Escarpment, summit (B1), 900 m, fr., 14 Feb 2022, Alade 574 (AUH); Idofin (A3), 760 m, st., 1 Nov 2024, Alade 879 (AUH).
Distribution. Grid squares A2, A3, B1.


1 2 3 4 5
A
B
C
D

Chapter 7

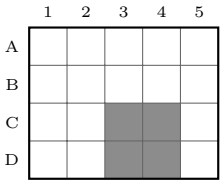
Irvingiaceae

Bush mango family

Trees. Leaves simple, alternate and entire. Stipules large and conical, enclosing the terminal bud and falling to leave a ring scar round the twig. Flowers small, 5-merous, in panicles. Fruit a large, fleshy or fibrous drupe. One species in the Arami Uplands.

7.1 *Irvingia gabonensis* (Aubry-Lecomte ex O'Rorke) Baill.

Synonyms. *Mangifera gabonensis* Aubry-Lecomte ex O'Rorke
Local and trade names. Yoruba: Oro; Igbo: Ogbono. Trade: Bush mango.
Status. Native.
Description. Evergreen tree to 40 m with buttresses. Bark grey-brown, smooth to finely scaly; slash yellowish. Leaves elliptic, 5–15 cm long, glossy above, with a short pointed tip. Terminal bud enclosed in a conical stipule 1–4 cm long. Flowers small, yellowish-green. Fruit an ellipsoid drupe 5–7 cm long, green turning yellow, with a sweet, fibrous pulp around a single woody stone.
Phenology. Flowering March to May; fruiting June to September.
Habitat. Lowland semi-deciduous forest.
Notes. Harris (1996) separated the bitter-fruited *Irvingia wombolu* from this species. It has not been recorded in the Arami Uplands.
Uses. The kernels are dried, ground and used to thicken soups (*ogbono* soup), and the sweet pulp is eaten fresh. The tree is increasingly planted in farms (Orwa et al., 2009).
Specimens examined. Olowo Stream (D3), 440 m, fr., 18 Jun 1934, Harding 1106 (OKF); Ireke Ferry (D4), 360 m, fr., 22 Jun 1936, Harding 1515 (OKF); Olowo Stream (D3), 440 m, fr., 22 Jun 1936, Harding 1524 (OKF); Igbo-Oba Forest Reserve (C3), 470 m, fr., 15 Jul 1937, Harding 1781 (OKF); Igbo-Oba Forest Reserve (C3), 470 m, st., 18 Jan 2020, Alade 246 (AUH); Okeri (C4), 420 m, st., 28 Jan 2021, Alade 445 (AUH); Olowo Stream (D3), 440 m, fr., 16 Aug 2021, Alade 538 (AUH).
Distribution. Grid squares C3, C4, D3, D4.



Chapter 8

Malvaceae

Mallow family

Trees, shrubs or herbs, often with star-shaped hairs. Leaves alternate, simple or digitately compound, with stipules. Flowers usually 5-merous, often large and showy. Fruit a capsule, a group of follicles, or indehiscent. The family is taken here in its broad modern sense, which includes the former Bombacaceae (*Adansonia*, *Ceiba*) and Sterculiaceae (*Cola*). Three species in the Arami Uplands.

Key to the species

1a Leaves simple; fruit of up to 5 spreading follicles with large seeds ..... *Cola nitida*

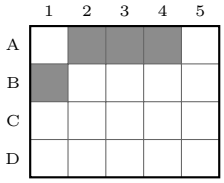
1b Leaves digitately compound 2

2a Trunk massive and swollen; flowers large, white and pendent; fruit with a woody shell and a dry, powdery pulp ..... *Adansonia digitata*

2b Trunk tall, with large buttresses and often spiny when young; fruit a capsule filled with silky floss *Ceiba pentandra*

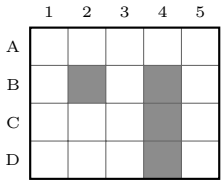
8.1 *Adansonia digitata* L.

Local and trade names. Yoruba: Ose; Hausa: Kuka. Trade: Baobab.
Status. Native.
Description. Deciduous tree to 20 m with a massive, swollen trunk often more than 3 m in diameter. Bark smooth, grey. Leaves digitately compound with 5–7 leaflets. Flowers large, white, pendent on long stalks, opening at night. Fruit ovoid to oblong, 15–25 cm long, with a velvety, woody shell and a whitish, powdery pulp.
Phenology. Flowering April to June; fruiting October to February.
Habitat. Guinea savanna woodland, often near old settlements.
Uses. The young leaves are cooked as a vegetable, the fruit pulp is made into a drink, and the bark fibre is twisted into rope (Burkill, 1985; Arbonnier, 2004).
Specimens examined. Idofin (A3), 760 m, fl., 1 May 1961, Olaniyan 325 (SVN); Okuta Hill (A4), 820 m, fr., 2 Dec 1962, Olaniyan 460 (SVN); Aramiwo Escarpment, summit (B1), 900 m, fr., 15 Dec 1962, Olaniyan 481 (SVN); Aramiwo Escarpment, summit (B1), 900 m, fr., 3 Nov 1967, Olaniyan 997 (SVN); Asiri Crossing (A2), 740 m, st., 18 Sep 2020, Alade 327 (AUH); Idofin (A3), 760 m, st., 10 Mar 2021, Alade 492 (AUH); Aramiwo Escarpment, summit (B1), 900 m, fl., 23 May 2023, Alade 771 (AUH).
Distribution. Grid squares A2, A3, A4, B1.



8.2 *Ceiba pentandra* (L.) Gaertn.

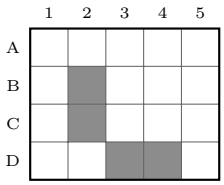
Synonyms. *Bombax pentandrum* L.
Local and trade names. Yoruba: Araba; Hausa: Rimi. Trade: Ceiba.
Status. Native.
Description. Deciduous tree to 50 m with large plank buttresses; young trunks and branches often spiny. Branches horizontal, in tiers. Leaves digitately compound with 5–9 leaflets. Flowers whitish to pale pink, appearing while the tree is leafless. Fruit an oblong capsule 10–25 cm long, filled with silky floss (kapok) round the seeds.
Phenology. Flowering December to February; fruiting February to April.
Habitat. Lowland semi-deciduous forest and forest–savanna mosaic; often left standing in villages and farmland.
Uses. The floss is used to stuff mattresses and pillows, and the light timber for canoes and boxes. Large trees are protected as sacred in many villages (Burkill, 1985; Keay, 1989).
Specimens examined. Ireke Ferry (D4), 360 m, fr., 3 Mar 1935, Harding 1353 (OKF); Ireke Ferry (D4), 360 m, fl., 20 Feb 1936, Harding 1415 (OKF); Okeri (C4), 420 m, fr., 6 Feb 1937, Harding 1616 (OKF); Aramiwo Foot (B2), 640 m, fr., 16 Mar 1963, Olaniyan 607 (SVN); Aramiwo Foot (B2), 640 m, fl., 7 Jan 1965, Olaniyan 697 (SVN); Oke-Asa farmland (B4), 690 m, fr., 6 Mar 2019, Alade 185 (AUH); Oke-Asa farmland (B4), 690 m, fr., 20 Apr 2023, Alade 719 (AUH).
Distribution. Grid squares B2, B4, C4, D4.



8.3 *Cola nitida* (Vent.) Schott & Endl.

Synonyms. *Sterculia nitida* Vent.
Local and trade names. Yoruba: Obi; Hausa: Goro. Trade: Kola.
Status. Cultivated. Native to the forests further west and widely planted in the Uplands.
Description. Evergreen tree to 20 m. Leaves simple, elliptic to obovate, 12–25 cm long, the petiole swollen at both ends. Flowers cream with a purplish-red centre. Fruit of up to 5 spreading follicles, each with several large red or white seeds (kola nuts).
Phenology. Flowering March to June; fruiting October to January.
Habitat. Farms and compounds in the lowland semi-deciduous forest and forest–savanna mosaic.
Uses. The nuts contain caffeine and are chewed as a stimulant. They are central to hospitality and ceremony and are traded widely across West Africa (Keay, 1989; Orwa et al., 2009).
Specimens examined. Ireke Ferry (D4), 360 m, fr., 9 Jan 1934, Harding 1049 (OKF); Ireke Ferry (D4), 360 m, fr., 11 Nov 1934, Harding 1236 (OKF); Olowo Stream (D3), 440 m, fl., 18 Jun 1938, Harding 1970 (OKF); Aramiwo Foot (B2), 640 m, fr., 2 Jan 1961, Olaniyan 247 (SVN); Aramiwo Foot (B2), 640 m, st., 10 Aug 1962, Olaniyan 450 (SVN); Aramiwo Foot (B2), 640 m, fl., 12 Jun 2020, Alade 282 (AUH); Agbede (C2), 540 m, st., 2 Aug 2020, Alade 297 (AUH).

Distribution. Grid squares B2, C2, D3, D4.



Chapter 9

Meliaceae

Mahogany family

Trees or shrubs, the bark and wood often scented. Leaves alternate, usually pinnate, without stipules. Flowers small, 4- or 5-merous, the stamens joined into a tube. Fruit a capsule, a berry or a drupe; in the timber genera treated here a woody capsule with winged seeds. Generic limits follow Pennington and Styles (1975). Three species in the Arami Uplands, all valuable timber trees.

Key to the species

1a Capsule cylindrical, pendulous, splitting from the base into 5 valves; slash with a cedar-like scent *Entandrophragma cylindricum*

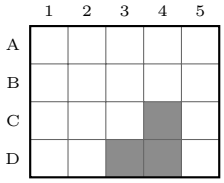
1b Capsule globose, erect, opening from the apex into 4 or 5 valves 2

2a Leaflets broadly obovate, up to 20 cm long; capsule 6–8 cm across; forest tree *Khaya grandifoliola*

2b Leaflets oblong-elliptic, up to 10 cm long; capsule 4–6 cm across; savanna tree *Khaya senegalensis*

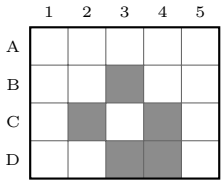
9.1 *Entandrophragma cylindricum* (Sprague) Sprague

Synonyms. *Pseudocedrela cylindrica* Sprague
Local and trade names. Trade: Sapele.
Status. Native.
Description. Deciduous tree to 50 m with short, blunt buttresses. Bark grey-brown, flaking in small round scales; slash with a cedar-like scent. Leaves paripinnate with 5–9 pairs of leaflets. Flowers small, greenish-yellow. Capsule cylindrical, pendulous, 6–10 cm long, splitting from the base into 5 valves; seeds winged.
Phenology. Flowering March to May; fruiting November to January.
Habitat. Lowland semi-deciduous forest.
Uses. A reddish-brown timber with a striped figure, used for veneer, furniture and joinery. Heavy logging has made large trees scarce (Hawthorne and Jongkind, 2006; Louppe et al., 2008).
Specimens examined. Ireke Ferry (D4), 360 m, fr., 10 Jan 1934, Harding 1059 (OKF); Ireke Ferry (D4), 360 m, st., 20 Sep 1934, Harding 1209 (OKF); Ireke Ferry (D4), 360 m, fr., 14 Dec 1936, Harding 1598 (OKF); Okeri (C4), 420 m, fr., 17 Dec 1938, Harding 2055 (OKF); Ireke Ferry (D4), 360 m, fr., 19 Jan 2019, Alade 145 (AUH); Olowo Stream (D3), 440 m, fr., 2 Jan 2024, Alade 840 (AUH); Olowo Stream (D3), 440 m, fr., 3 Dec 2024, Alade 938 (AUH).
Distribution. Grid squares C4, D3, D4.



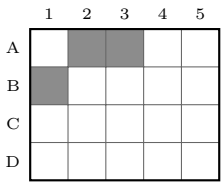
9.2 *Khaya grandifoliola* C.DC.

Local and trade names. Yoruba: Oganwo. Trade: African mahogany.
Status. Native.
Description. Deciduous tree to 40 m with buttresses. Bark grey-brown, flaking in scales; slash red. Leaves paripinnate with 2–5 pairs of broadly obovate leaflets up to 20 cm long. Flowers small, cream. Capsule erect, globose, 6–8 cm across, opening into 4 or 5 valves; seeds winged.
Phenology. Flowering March to May; fruiting January to March.
Habitat. Lowland semi-deciduous forest and forest–savanna mosaic.
Uses. A valuable mahogany timber for furniture and joinery. The bitter bark is used in local medicine (Keay, 1989; Louppe et al., 2008).
Specimens examined. Ireke Ferry (D4), 360 m, fr., 28 Jan 1936, Harding 1391 (OKF); Olowo Stream (D3), 440 m, st., 6 Nov 1937, Harding 1805 (OKF); Ireke Ferry (D4), 360 m, fr., 20 Mar 1938, Harding 1868 (OKF); Ewe Stream gallery forest (B3), 580 m, fr., 10 Feb 1961, Olaniyan 293 (SVN); Ewe Stream gallery forest (B3), 580 m, fr., 8 Feb 1966, Olaniyan 774 (SVN); Okeri (C4), 420 m, fr., 15 Mar 2019, Alade 189 (AUH); Agbede (C2), 540 m, fr., 26 Mar 2020, Alade 263 (AUH).
Distribution. Grid squares B3, C2, C4, D3, D4.



9.3 *Khaya senegalensis* (Desr.) A.Juss.

Synonyms. *Swietenia senegalensis* Desr.
Local and trade names. Hausa: Madachi. Trade: Dry-zone mahogany.
Status. Native.
Description. Tree to 30 m with a dense crown. Bark dark grey, scaly. Leaves paripinnate with 3–6 pairs of oblong-elliptic leaflets up to 10 cm long. Flowers small, cream. Capsule erect, globose, 4–6 cm across, opening into 4 valves; seeds winged.
Phenology. Flowering March to May; fruiting January to February.
Habitat. Guinea savanna woodland, especially along watercourses; also planted as a shade and avenue tree.
Uses. Timber for furniture and building. The very bitter bark is used in local medicine, and the branches are lopped for fodder in the dry season (Arbonnier, 2004; Orwa et al., 2009).
Specimens examined. Aramiwo Escarpment, summit (B1), 900 m, fr., 22 Feb 1962, Olaniyan 360 (SVN); Idofin (A3), 760 m, fr., 14 Jan 1963, Olaniyan 515 (SVN); Aramiwo Escarpment, summit (B1), 900 m, fr., 26 Jan 1967, Olaniyan 878 (SVN); Aramiwo Escarpment, summit (B1), 900 m, fr., 13 Feb 1967, Olaniyan 909 (SVN); Idofin (A3), 760 m, fl., 10 Apr 2019, Alade 207 (AUH); Aramiwo Escarpment, summit (B1), 900 m, fl., 6 Mar 2023, Alade 685 (AUH); Asiri Crossing (A2), 740 m, fr., 22 Jan 2025, Alade 974 (AUH).
Distribution. Grid squares A2, A3, B1.


1 2 3 4 5
A
B
C
D

Chapter 10

Moraceae

Fig family

Trees, shrubs or climbers with white latex. Leaves alternate, simple, with stipules that fall early and leave a scar round the twig. Flowers small and unisexual, crowded in heads, spikes or catkins, or enclosed in a fleshy receptacle. Fruits often fused into a fleshy compound fruit. Two species in the Arami Uplands, both timber trees.

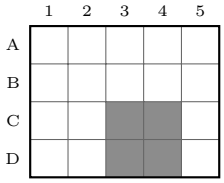
Key to the species

1a Leaves unequal at the base, rough; male flowers in small, flat heads; fruit a single red, velvety drupe *Antiaris toxicaria*

1b Leaves equal at the base, smooth; flowers in catkins, the male catkins long and pendent; fruit a fleshy green compound fruit like a large mulberry ..... *Milicia excelsa*

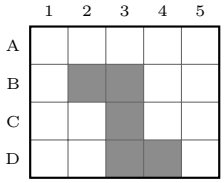
10.1 *Antiaris toxicaria* Lesch.

Synonyms. *Antiaris africana* Engl.
Local and trade names. Trade: Ako.
Status. Native.
Description. Deciduous tree to 40 m with small buttresses. Bark smooth, grey to whitish, exuding abundant white latex when cut. Leaves oblong-elliptic, 8–20 cm long, unequal at the base and rough to the touch, especially on young trees. Male flowers in small, flat heads; female flowers solitary. Fruit a red, velvety drupe about 2 cm long.
Phenology. Flowering November to February; fruiting February to April.
Habitat. Lowland semi-deciduous forest.
Notes. West African plants are referred to *Antiaris toxicaria* subsp. *welwitschii* (Engl.) C.C.Berg.
Uses. A light timber used for plywood and boxes. The inner bark has been beaten into bark cloth (Hawthorne and Jongkind, 2006; Louppe et al., 2008).
Specimens examined. Olowo Stream (D3), 440 m, st., 18 Jul 1934, Harding 1181 (OKF); Ireke Ferry (D4), 360 m, fr., 23 Feb 1937, Harding 1630 (OKF); Olowo Stream (D3), 440 m, fr., 7 Mar 1938, Harding 1847 (OKF); Okeri (C4), 420 m, fr., 16 Apr 1938, Harding 1927 (OKF); Igbo-Oba Forest Reserve (C3), 470 m, fr., 5 Feb 2021, Alade 451 (AUH); Okeri (C4), 420 m, fr., 5 Feb 2021, Alade 458 (AUH); Ireke Ferry (D4), 360 m, fl., 17 Nov 2024, Alade 934 (AUH).
Distribution. Grid squares C3, C4, D3, D4.



10.2 *Milicia excelsa* (Welw.) C.C.Berg

Synonyms. *Morus excelsa* Welw.
Local and trade names. Yoruba: Iroko. Trade: Iroko.
Status. Native.
Description. Deciduous tree to 50 m with a long, clear bole and small buttresses. Bark pale grey, scaly, exuding white latex. Leaves elliptic, 10–20 cm long, pointed at the tip and rounded at the base. Male and female flowers on separate trees; male catkins 10–20 cm long, pendent and white; female catkins shorter and green. Fruit a fleshy, green compound fruit 5–7 cm long, like a large mulberry.
Phenology. Flowering October to November; fruiting December to February.
Habitat. Lowland semi-deciduous forest and forest–savanna mosaic; large trees are often left standing in farmland.
Notes. This species was long placed in *Chlorophora*; Berg (1982) reinstated the genus *Milicia* for it.
Uses. One of the most important timbers of West Africa, very durable and used for building, furniture, flooring and boats. Old trees are held sacred in many communities (Keay, 1989; Louppe et al., 2008).
Specimens examined. Ireke Ferry (D4), 360 m, st., 20 May 1936, Harding 1452 (OKF); Igbo-Oba Forest Reserve (C3), 470 m, fr., 16 Jan 1938, Harding 1833 (OKF); Olowo Stream (D3), 440 m, fl., 28 Nov 1938, Harding 2025 (OKF); Aramiwo Foot (B2), 640 m, fl., 21 Oct 1961, Olaniyan 338 (SVN); Ewe Stream gallery forest (B3), 580 m, st., 15 Mar 1964, Olaniyan 677 (SVN); Igbo-Oba Forest Reserve (C3), 470 m, fl., 17 Oct 2021, Alade 548 (AUH); Igbo-Oba Forest Reserve (C3), 470 m, fr., 2 Jan 2024, Alade 858 (AUH).
Distribution. Grid squares B2, B3, C3, D3, D4.



Chapter 11

Sapotaceae

Shea family

Trees or shrubs with white latex. Leaves alternate, simple and entire, often crowded at the ends of the twigs, without stipules, often with silky or rusty hairs beneath. Flowers small, in clusters in the leaf axils or at the ends of the twigs. Fruit a berry with one to several large, glossy seeds, each with a broad scar. Two species in the Arami Uplands, both valued for their fruits.

Key to the species

1a Leaves whitish beneath; fruit orange-yellow with 4 or 5 glossy seeds; forest tree .. *Chrysophyllum albidum*

1b Leaves hairless beneath, crowded at the ends of stout twigs; bark thick and corky, cracked into blocks; fruit green to yellowish with usually one seed; savanna tree ..... *Vitellaria paradoxa*

11.1 *Chrysophyllum albidum* G.Don

Local and trade names. Yoruba: Agbalumo; Igbo: Udara. Trade: African star apple.

Status. Native.

Description. Evergreen tree to 35 m with a bole fluted at the base. Bark grey-brown; slash with white latex. Leaves elliptic to oblong, 8–20 cm long, whitish beneath. Flowers small, cream, clustered in the leaf axils. Fruit a globose berry 3–5 cm across, orange-yellow when ripe, with a sweet-sour pulp and 4 or 5 glossy brown seeds.

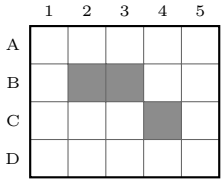
Phenology. Flowering March to May; fruiting December to March.

Habitat. Lowland semi-deciduous forest and forest–savanna mosaic; often kept or planted near houses.

Uses. The fruit is eaten fresh and sold in large quantities in markets during the dry season (Keay, 1989; Orwa et al., 2009).

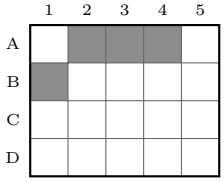
Specimens examined. Okeri (C4), 420 m, fr., 23 Dec 1934, Harding 1287 (OKF); Okeri (C4), 420 m, fl., 11 Apr 1936, Harding 1423 (OKF); Okeri (C4), 420 m, fl., 26 May 1936, Harding 1457 (OKF); Aramiwo Foot (B2), 640 m, fr., 10 Mar 1963, Olaniyan 595 (SVN); Ewe Stream gallery forest (B3), 580 m, fl., 18 May 1965, Olaniyan 719 (SVN); Aramiwo Foot (B2), 640 m, fr., 7 Dec 2020, Alade 360 (AUH); Okeri (C4), 420 m, fr., 14 Mar 2023, Alade 690 (AUH).

Distribution. Grid squares B2, B3, C4.



11.2 *Vitellaria paradoxa* C.F.Gaertn.

Local and trade names. Yoruba: Emi; Hausa: Kadanya. Trade: Shea.
Status. Native.
Description. Deciduous tree to 15 m. Bark dark grey, thick and corky, cracked into rectangular blocks. Twigs stout. Leaves oblong, 10–25 cm long, crowded at the ends of the twigs, with wavy margins. Flowers creamy white, fragrant, clustered at the ends of the twigs, appearing while the tree is leafless. Fruit an ellipsoid berry 4–5 cm long, green to yellowish, with a sweet pulp and usually one large seed (the shea nut).
Phenology. Flowering January to March; fruiting May to July.
Habitat. Guinea savanna woodland; kept in farmland when land is cleared, forming open parkland.
Notes. This species has often been placed in the genus *Butyrospermum*.
Uses. Shea butter, pressed from the kernels, is used for cooking and in cosmetics and is an important export. The fruit pulp is eaten (Arbonnier, 2004; Orwa et al., 2009).
Specimens examined. Asiri Crossing (A2), 740 m, fl., 15 Mar 1963, Olaniyan 599 (SVN); Aramiwo Escarpment, summit (B1), 900 m, fr., 4 Jun 1963, Olaniyan 628 (SVN); Okuta Hill (A4), 820 m, fr., 4 Jun 1964, Olaniyan 687 (SVN); Aramiwo Escarpment, summit (B1), 900 m, fr., 7 Jul 1966, Olaniyan 826 (SVN); Aramiwo Escarpment, summit (B1), 900 m, fr., 23 Jul 2019, Alade 226 (AUH); Idofin (A3), 760 m, st., 15 Apr 2021, Alade 504 (AUH); Asiri Crossing (A2), 740 m, fr., 17 Jul 2025, Alade 984 (AUH).
Distribution. Grid squares A2, A3, A4, B1.



Appendix A

Author Abbreviations

Author abbreviations used in the names in this volume follow Brummitt and Powell (1992). Every abbreviation that appears in an author citation in this volume must be listed here.

Abbreviation	Author	Dates
A.Chev.	Auguste Jean Baptiste Chevalier	1873–1956
A.Juss.	Adrien-Henri de Jussieu	1797–1853
Aubry-Lecomte	Charles Eugène Aubry-Lecomte	1821–1898
Baill.	Henri Ernest Baillon	1827–1895
C.C.Berg	Cornelis Christiaan Berg	1934–2012
C.DC.	Anne Casimir Pyramus de Candolle	1836–1918
C.F.Gaertn.	Carl Friedrich von Gaertner	1772–1850
Desr.	Louis Auguste Joseph Desrousseaux	1753–1838
Diels	Friedrich Ludwig Emil Diels	1874–1945
Endl.	Stephan Friedrich Ladislaus Endlicher	1804–1849
Engl.	Heinrich Gustav Adolf Engler	1844–1930
G.Don	George Don	1798–1856
Gaertn.	Joseph Gaertner	1732–1791
Jacq.	Nikolaus Joseph von Jacquin	1727–1817
L.	Carl Linnaeus	1707–1778
Lesch.	Jean Baptiste Louis Théodore Leschenault de La Tour	1773–1826
O'Rorke	O'Rorke	not traced
Pers.	Christiaan Hendrik Persoon	1761–1836
Poir.	Jean Louis Marie Poiret	1755–1834
R.Br.	Robert Brown	1773–1858
Schott	Heinrich Wilhelm Schott	1794–1865
Sm.	James Edward Smith	1759–1828
Sprague	Thomas Archibald Sprague	1877–1958
Vent.	Étienne Pierre Ventenat	1757–1808
Welw.	Friedrich Martin Josef Welwitsch	1806–1872

Appendix B

Gazetteer of Localities

This gazetteer lists every locality cited in the specimen records of this volume. For reference, the Flora area is divided into grid squares: rows A to D run from north to south, and columns 1 to 5 run from west to east. Altitudes are in metres. Vegetation types follow the Introduction: LSF, lowland semi-deciduous forest (below 500 m); FSM, forest–savanna mosaic (500–700 m); GSW, Guinea savanna woodland (above 700 m).

Locality	Grid square	Altitude (m)	Vegetation
Agbede	C2	540	FSM
Aramiwo Escarpment, summit	B1	900	GSW
Aramiwo Foot	B2	640	FSM
Asiri Crossing	A2	740	GSW
Ewe Stream gallery forest	B3	580	FSM
Idofin	A3	760	GSW
Igbo-Oba Forest Reserve	C3	470	LSF
Ireke Ferry	D4	360	LSF
Oke-Asa farmland	B4	690	FSM
Okeri	C4	420	LSF
Okuta Hill	A4	820	GSW
Olowo Stream	D3	440	LSF

Appendix C

Phenological Calendar

The calendar summarises `data/phenology.tsv`. F, recorded in flower; R, recorded in fruit; F+R, both; –, neither. It agrees month for month with the phenology statement in each species treatment.

Table C.1: Months in which each species is recorded in flower or fruit.

Species	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
A. occidentale	F	F+R	F+R	R	R	–	–	–	–	–	–	F
S. mombin	–	F	F	F	–	R	R	R	R	–	–	–
T. ivorensis	–	F	F	F+R	R	R	–	–	–	–	–	–
T. superba	–	F	F	F	R	R	R	R	–	–	–	–
A. africana	R	R	F	F	F	–	–	–	–	–	R	R
P. biglobosa	F	F	R	R	R	–	–	–	–	–	–	F
P. erinaceus	F	F+R	R	R	–	–	–	–	–	–	–	F
I. gabonensis	–	–	F	F	F	R	R	R	R	–	–	–
A. digitata	R	R	–	F	F	F	–	–	–	R	R	R
C. pentandra	F	F+R	R	R	–	–	–	–	–	–	–	F
C. nitida	R	–	F	F	F	F	–	–	–	R	R	R
E. cylindricum	R	–	F	F	F	–	–	–	–	–	R	R
K. grandifoliola	R	R	F+R	F	F	–	–	–	–	–	–	–
K. senegalensis	R	R	F	F	F	–	–	–	–	–	–	–
A. toxicaria	F	F+R	R	R	–	–	–	–	–	–	F	F
M. excelsa	R	R	–	–	–	–	–	–	–	F	F	R
C. albidum	R	R	F+R	F	F	–	–	–	–	–	–	R
V. paradoxa	F	F	F	–	R	R	R	–	–	–	–	–

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